

Qualitative Coding Report: Scientific Communities in the Literature Selection

Basis: 250 records from the uploaded CSV file. Coding uses only the columns "title" and "keywords".

Purpose: identify the scientific communities represented in the literature selection through inductive content categories.

Methodological note

I treated titles and keywords as short qualitative text segments. I first inspected repeated terms and topical clusters, then consolidated them into broader communities. Because individual papers can connect more than one community, the report distinguishes conceptual overlap from a single primary code used only for the count table. The primary code reflects the strongest title/keyword signal.

The categories are interpretive and content-based rather than journal-based. They should therefore be read as communities of topic, method, and discourse visible in the literature selection.

Overview of primary coding

Community / category	Primary n	Share	Mention n	Definition
Assessment, Measurement, Psychometrics and Evaluation	37	14.8%	83	Research communities concerned with educational measurement, test development, validation, scoring, survey instruments, program evaluation, and the interpretation of assessment results.
STEM, Science and Mathematics Education	26	10.4%	44	Communities studying teaching and learning in science, technology, engineering, mathematics, and related disciplinary education contexts, including science communication and STEM innovation.
Teacher Education, Professional Learning and Teaching Practice	25	10.0%	64	Research on teachers, instructors, teacher education, professional development, pedagogical practice, and professional learning networks.
COVID-19, Well-being and Socio-Emotional Learning	24	9.6%	37	Communities focusing on pandemic-related education, student and teacher well-being, mental health, motivation, emotions, stress, self-regulation, and socio-emotional outcomes.
Reading, Literacy, Language and Curriculum Studies	23	9.2%	44	Communities studying reading, literacy, language learning, writing, textbooks, curriculum, content analysis, and textual or curricular representations.
AI, Educational Data Science and NLP	23	9.2%	32	Communities using or studying artificial intelligence, machine learning, natural language processing, data science, text mining, explainable models, and AI-based educational applications.
Learning Analytics, Online and Technology-Enhanced Learning	17	6.8%	54	Research on digital learning environments, learning analytics, online and blended education, ICT, educational technology, virtual or augmented reality, platforms, and digital interventions.
Higher Education, Graduate Education and Academic Careers	17	6.8%	52	Communities focused on postsecondary, undergraduate, graduate, and doctoral education, student success in college, academic careers, curricula, and career outcomes.
Equity, Inclusion, Diversity and Special Education	16	6.4%	43	Research communities investigating inclusion, access, representation, disability, special educational needs, race, gender, socioeconomic inequalities, gifted education, and educational justice.
Open Science, Bibliometrics and Research Infrastructure	15	6.0%	27	Communities concerned with open science practices, reproducibility, data sharing, preregistration, bibliometrics, systematic/scoping reviews, research mapping, and research infrastructure.
Other / Cross-disciplinary Education Research	14	5.6%	0	A residual cross-disciplinary community for education studies whose titles and keywords do not clearly map to one of the dominant communities above or combine several themes without a single clear center.
Academic Integrity, Ethics and Responsible AI	7	2.8%	16	Communities studying academic integrity, cheating, plagiarism, online proctoring, AI detection, ethical perspectives on generative AI, privacy, bias, trust, and responsible AI use.
Early Childhood, Primary School and Child Development	6	2.4%	29	Communities focused on early childhood education, primary or elementary schooling, child development, working memory, self-regulation, and early learning environments.

Category definitions and examples

Assessment, Measurement, Psychometrics and Evaluation

Primary coded records: 37 of 250. Any mention: 83 records.

Definition: Research communities concerned with educational measurement, test development, validation, scoring, survey instruments, program evaluation, and the interpretation of assessment results.

Typical title/keyword signals: assessment; measurement; psychometrics; validation; Rasch; test; evaluation; survey; response scales.

Example: Development and validation of a rapid and precise online sentence reading efficiency assessment.

Example keywords: dyslexia; reading fluency assessment; screening tools; psychometrics.

STEM, Science and Mathematics Education

Primary coded records: 26 of 250. Any mention: 44 records.

Definition: Communities studying teaching and learning in science, technology, engineering, mathematics, and related disciplinary education contexts, including science communication and STEM innovation.

Typical title/keyword signals: STEM; mathematics; science education; engineering; computational thinking; NGSS; science communication.

Example: Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database.

Example keywords: ICT; mathematics education; bibliometric analysis; Scopus.

Teacher Education, Professional Learning and Teaching Practice

Primary coded records: 25 of 250. Any mention: 64 records.

Definition: Research on teachers, instructors, teacher education, professional development, pedagogical practice, and professional learning networks.

Typical title/keyword signals: teacher; teaching; teacher education; professional development; pedagogical; classroom practice; instructor.

Example: A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic.

Example keywords: teacher learning; professional learning networks; COVID-19; social media.

COVID-19, Well-being and Socio-Emotional Learning

Primary coded records: 24 of 250. Any mention: 37 records.

Definition: Communities focusing on pandemic-related education, student and teacher well-being, mental health, motivation, emotions, stress, self-regulation, and socio-emotional outcomes.

Typical title/keyword signals: COVID-19; pandemic; lockdown; well-being; mental health; stress; emotions; motivation; self-regulation.

Example: Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic.

Example keywords: COVID-19 pandemic; college students; psychological well-being; emotions; personality.

Reading, Literacy, Language and Curriculum Studies

Primary coded records: 23 of 250. Any mention: 44 records.

Definition: Communities studying reading, literacy, language learning, writing, textbooks, curriculum, content analysis, and textual or curricular representations.

Typical title/keyword signals: reading; literacy; language; textbook; curriculum; writing; comprehension; fluency; bilingualism.

Example: Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks.

Example keywords: content analysis; curriculum; gender studies; history; race; textbooks; textual analysis.

AI, Educational Data Science and NLP

Primary coded records: 23 of 250. Any mention: 32 records.

Definition: Communities using or studying artificial intelligence, machine learning, natural language processing, data science, text mining, explainable models, and AI-based educational applications.

Typical title/keyword signals: AI; artificial intelligence; machine learning; NLP; data science; text analysis; sentiment analysis; GPT.

Example: Are U.S. graduate curricula ready to prepare social data scientists for the AI era?.

Example keywords: artificial intelligence; coursework; curriculum gap; data science; Rasch modeling; web-scraping.

Learning Analytics, Online and Technology-Enhanced Learning

Primary coded records: 17 of 250. Any mention: 54 records.

Definition: Research on digital learning environments, learning analytics, online and blended education, ICT, educational technology, virtual or augmented reality, platforms, and digital interventions.

Typical title/keyword signals: learning analytics; online learning; e-learning; technology-enhanced learning; digital tools; ICT; VR; AR.

Example: A longitudinal study of interplay between student engagement and self-regulation.

Example keywords: learning analytics; self-regulated learning; online education; sequence analysis; hidden Markov models.

Higher Education, Graduate Education and Academic Careers

Primary coded records: 17 of 250. Any mention: 52 records.

Definition: Communities focused on postsecondary, undergraduate, graduate, and doctoral education, student success in college, academic careers, curricula, and career outcomes.

Typical title/keyword signals: higher education; college; university; undergraduate; graduate; doctoral; PhD; postsecondary; career outcomes.

Example: Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education.

Example keywords: graduate education; doctoral training; program evaluation; career outcomes; taxonomy; PhD.

Equity, Inclusion, Diversity and Special Education

Primary coded records: 16 of 250. Any mention: 43 records.

Definition: Research communities investigating inclusion, access, representation, disability, special educational needs, race, gender, socioeconomic inequalities, gifted education, and educational justice.

Typical title/keyword signals: equity; inclusion; diversity; race; ethnicity; gender; first-generation; special education; disability; access.

Example: Going to College With a Posse: How Having High School Peers on Campus Supports College Achievement.

Example keywords: equity; first-generation college students; privilege; social belonging; socioeconomic status.

Open Science, Bibliometrics and Research Infrastructure

Primary coded records: 15 of 250. Any mention: 27 records.

Definition: Communities concerned with open science practices, reproducibility, data sharing, preregistration, bibliometrics, systematic/scoping reviews, research mapping, and research infrastructure.

Typical title/keyword signals: open science; open data; preregistration; data sharing; bibliometric; systematic review; scoping review; meta-analysis.

Example: Data Sharing in Education Science.

Example keywords: data sharing; education science; open science; research integrity; reproducibility; data repositories.

Other / Cross-disciplinary Education Research

Primary coded records: 14 of 250. Any mention: 0 records.

Definition: A residual cross-disciplinary community for education studies whose titles and keywords do not clearly map to one of the dominant communities above or combine several themes without a single clear center.

Typical title/keyword signals: general education research; mixed or broad themes; learning situations; policy contexts; cross-cutting studies.

Example: Disentangling objective characteristics of learning situations from subjective perceptions thereof; using an experience sampling method design.

Example keywords: experience sampling method; learning situations; subjective perceptions; situational interest.

Academic Integrity, Ethics and Responsible AI

Primary coded records: 7 of 250. Any mention: 16 records.

Definition: Communities studying academic integrity, cheating, plagiarism, online proctoring, AI detection, ethical perspectives on generative AI, privacy, bias, trust, and responsible AI use.

Typical title/keyword signals: academic integrity; cheating; plagiarism; proctoring; AI detection; ethics; responsible AI; privacy; bias; trust.

Example: Academic Integrity in Online Assessment: A Research Review.

Example keywords: academic integrity; academic dishonesty; cheating; online courses; remote teaching.

Early Childhood, Primary School and Child Development

Primary coded records: 6 of 250. Any mention: 29 records.

Definition: Communities focused on early childhood education, primary or elementary schooling, child development, working memory, self-regulation, and early learning environments.

Typical title/keyword signals: early childhood; children; kindergarten; preschool; elementary; primary school; working memory; developmental.

Example: Determine the impact of Emotive Intelligent Spaces on childrens behavioural and cognitive outcomes.

Example keywords: early childhood; colour; intervention; working memory; self-regulation.

Interpretive synthesis

The strongest visible communities are measurement/evaluation, STEM/science/mathematics education, teacher professional learning, COVID-19 and well-being, reading/literacy/curriculum studies, and AI/data science. This indicates a literature selection situated at the intersection of empirical education research, educational technology, assessment science, and open or computational research practices.

Several communities overlap substantially. For example, AI studies often also appear in assessment, academic integrity, inclusive education, or curriculum analysis. Similarly, STEM papers frequently overlap with technology-enhanced learning and open-science or bibliometric review work. The primary counts therefore summarize the dominant signal, not exclusive membership.

For further manual refinement, I would recommend double-coding records that contain both AI/data-science and domain-specific terms, because these papers may represent methodological communities as much as substantive education communities.

Reproducibility note

The analysis was generated from the uploaded CSV using only the title and keywords fields. A primary code was assigned to each record for counting; the PDF reports the conceptual categories, definitions, indicative terms, and one example per category.